

UrsAi (Atlas AI) Platform

End-to-End Complete Project Structure & Architectural Specification Report

Project Name:	UrsAi / Atlas AI (UrsBiz)	Version:	1.0.0 (Release Candidate)
Repository:	vishwanathbs03/UrsAii	Target Industry:	MSME Intelligence & Growth Engine
Frontend Tech:	Next.js 14, React 18, TypeScript, Tailwind	Backend Tech:	FastAPI (Python 3.12), SQLAlchemy, SQLite
AI & Decision Core:	Business DNA, Digital Twin, Rule Engine	Deployment:	Docker Compose, Nginx, Health Checkers

1. Executive Summary & Core Platform Vision

UrsAi (Atlas AI) is an enterprise-grade AI-powered Business Intelligence and Digital Twin simulation platform designed specifically for Micro, Small, and Medium Enterprises (MSMEs). The platform provides end-to-end operational analysis, synthesizing multi-dimensional business inputs (financials, workforce, market presence, digital readiness, compliance) into a living **Digital Twin**. Powered by a deterministic **Rule Engine**, a **Business DNA Profiler**, and predictive machine learning models, UrsAi generates actionable growth roadmaps, financial health scores (Business MRI), scheme eligibility analysis, and real-time risk mitigation strategies.

Core System Capabilities:

- **Digital Twin & Simulation:** Real-time state replication enabling 'what-if' scenario testing for investments, hiring, and expansion.
- **Business MRI & Scoring:** Multi-variable health scoring across 6 key dimensions: Financial, Operations, Digital, Compliance, Market, and Risk.
- **Automated Compliance & Schemes:** Real-time matching against government subsidies, MSME loan schemes, and regulatory requirements.
- **AI Copilot & Assistant:** Context-aware LLM & RAG-driven advisory assistant providing tailored strategic recommendations.
- **Predictive AI Forecasting:** Revenue, cash flow, and market expansion projections driven by historical metrics and industry benchmarks.
- **Action Board & Execution:** Interactive Kanban management for executing recommended strategic initiatives with tracked ROI.

2. High-Level System Architecture & Stack Overview

The project strictly adheres to a decoupled client-server architecture. The Next.js frontend communicates with the FastAPI REST backend via structured JSON payloads, protected by JWT authentication and middleware layers.

Layer / Subsystem	Technology Stack & Libraries	Key Responsibilities
Frontend Web App	Next.js 14 (App Router), React 18, TypeScript, TailwindCSS, Lucide Icons, Recharts, Framer Motion	Responsive UI, interactive dashboards, radar charts, Kanban boards, wizard forms, client state management.
Backend API Service	FastAPI, Python 3.12, Pydantic v2, Gunicorn, Uvicorn, Python-Dotenv, ReportLab	REST API routing (30+ endpoints), JWT authentication, request validation, business logic, PDF generation.
AI & Decision Core	Business DNA Engine, Digital Twin Engine, Rule Registry, Scenario Delta Calculator, RAG Retriever	Deterministic rule evaluation, Archetype scoring, SWOT extraction, what-if scenario simulations, knowledge retrieval.

Database & Storage	SQLite (atlas_ai.db), SQLAlchemy ORM, Alembic Migrations	Persistent storage for users, business profiles, goals, actions, chat logs, digital presence, and export history.
Deployment & Infra	Docker, Docker Compose, Nginx Reverse Proxy, Health Checkers	Containerized microservice orchestration, SSL termination, static file serving, zero-downtime reloads.

3. End-to-End Directory Tree Breakdown

Below is the complete, detailed directory hierarchy of the entire UrsAi codebase spanning root, backend, frontend, database, deployment, scripts, and documentation modules.

3.1 Root Directory (^/) Structure

File / Folder	Description & Role in Project
frontend/	Next.js 14 TypeScript web frontend application
backend/	FastAPI Python REST API backend and AI engine suite
database/	Database schema specifications and documentation
deployment/	Production deployment configurations and environment templates
docs/	Comprehensive engineering specifications (51 markdown documents)
scripts/	Automated sprint verification and behavioral test suites
verify_out/	Cached output fixtures for verification testing
docker-compose.yml	Docker Compose definition for local/dev multi-container setup
nginx.conf	Nginx reverse proxy configuration for frontend & backend routing
generate_pptx.py	Python presentation deck builder script for pitch decks
biz.json, twin.json...	Data fixtures for business DNA, digital twin, recommendations, and roadmaps
FINAL_PROJECT_REPORT.md	Final executive project completion and status summary

3.2 Backend Module (/backend/app) Detailed Architecture

The backend is structured into domain-driven packages inside backend/app/. Each domain contains API endpoints, Pydantic schemas, ORM models, data repositories, and specialized services.

Subdirectory	Module Name	Detailed Technical Description
api/v1/endpoints/	FastAPI Routes (30+ modules)	Exposes REST endpoints for auth, business, digital twin, DNA, dashboard, advisor, copilot, predictive analytics, schemes, OCR, risks, recommendations, and reports.
services/ai/	AI Orchestration & LLM	Includes context builders, prompt builders, response parsers, and multi-provider LLM integration (Ollama / Mock / Cloud LLM).
services/dna/	Business DNA Engine	Calculates MSME business DNA traits, archetype classification, confidence scores, SWOT matrices, and signal extraction.
services/twin/	Digital Twin Engine	Aggregates business state, health scores, risk matrices, opportunity matrices, historical snapshots, and timeline projections.
services/rules/	Rule Engine & Compliance	Evaluates operational rules across 9 priorities (compliance, digital, export, risk, immediate, long-term, high/medium priority).
services/scenario/	Scenario Simulator	Executes what-if simulation clones, applies mutations, computes state deltas, and calculates financial/operational impact.

services/scoring/	Business MRI Scoring	Computes multi-dimensional health scores, maturity levels, lifted factors, and derived radar dimensions.
services/finance/	Finance & Valuation	Calculates cash flow projections, funding eligibility, ROI metrics, valuation benchmarks, and financial summaries.
services/ocr/	OCR & Document Parser	Extracts structured business data from uploaded tax filings, financial statements, and compliance certificates.
models/ & schemas/	Data Contracts & ORM	SQLAlchemy database models and Pydantic v2 validation schemas for all data models (User, Business, ActionItem, Product, etc.).
middleware/ & monitoring/	Security & Monitoring	JWT authentication, CORS policy, rate limiting, performance latency tracking, health check endpoints, and structured logging.

3.3 Frontend Module (`/frontend`) Detailed Structure

The frontend is organized using Next.js 14 App Router, feature-based modular folders, reusable UI components, and TypeScript API service abstractions.

Directory Path	Component / Domain	Function & Page Description
app/	Next.js App Routes	Page routes including (auth)/, dashboard/, business/, simulation/, reports/, and settings/.
features/dashboard/	Dashboard Feature Suite	HealthScoreCard, ReadinessCard, DnaDimensions, ScoresRadarCard, SwotSummaryCard, KpiStrip, RecentActivityCard, DashboardView.
features/action-board/	Action Board & Kanban	Interactive Kanban board (KanbanColumn, ActionCard, ActionDetailsPanel, BoardSummaryPanel) for strategic execution.
features/advisor/	Strategic Advisor UI	AdvisorView, AdvisorSummaryCard, FundingCard, ComplianceCard, RiskCards, RecommendationCards, GrowthTips.
features/analytics/	Analytics & Radar Charts	MaturityRadarChart, ScoreTrendsChart, SchemeEligibilityChart, ScenarioSimulator, OpportunityAnalytics, RiskAnalytics.
features/assistant/	AI Assistant & RAG Chat	Conversational AI interface (MessageBubble, PromptInput, ConversationList, ContextPanel, SuggestedQuestions).
features/predictive-analytics/	Predictive Analytics	GrowthForecast, TimelineVisualization, ProjectionCards, WhatDrivesGrowth, PredictiveAnalyticsView.
features/reports/	Reports & Export	14 specialized report section components (BusinessDnaSection, RiskSummarySection, ExecutiveSummarySection) with PDF export.
components/ui/ & common/	Design System	Reusable UI components (button, input, password-input, toast, alert, form-field, Navbar, Sidebar, PageHeader, Footer).
services/	API Service Layer	TypeScript services (api-client.ts, auth-service.ts, business-service.ts, twin-service.ts, rules-service.ts, etc.) managing backend API integration.

4. End-to-End Data Flow & Lifecycle

Understanding how data moves through UrsAi from user input to executive decision output is critical to appreciating the architecture. The diagram and steps below articulate the complete processing lifecycle:

- **Step 1: Business Profile Onboarding:** User enters company data, financial metrics, workforce details, and digital footprint via the Business Wizard or OCR Document Upload.
- **Step 2: Business DNA Profiling & Archetype Extraction:** The services/dna/ engine analyzes raw traits to compute confidence scores, assign an MSME Archetype, and extract a dynamic SWOT matrix.
- **Step 3: Business MRI Health Scoring:** The services/scoring/ engine calculates overall health (0-100) across Financial, Operational, Digital, Risk, and Market dimensions.

- **Step 4: Rule Engine & Schemes Matching:** The services/rules/ engine evaluates rules across compliance, high-priority alerts, and matches business attributes with government schemes.
- **Step 5: Digital Twin Synthesis & Scenario Simulation:** State snapshot is aggregated in services/twin/. User can trigger 'what-if' simulations via services/scenario/ to test revenue/cost changes.
- **Step 6: Strategic Recommendations & Action Roadmap:** Priority recommendations and phased implementation roadmaps are generated and formatted into Kanban Action Boards.
- **Step 7: PDF Report & Dashboard Visualization:** Next.js frontend renders rich charts, radar visualizations, and backend ReportLab generates exportable executive PDF reports.

5. Engineering Documentation Catalog (`/docs`)

The project features 51 detailed markdown specifications inside /docs establishing strict development guidelines, architecture principles, and system requirements. Key documents include:

Specification File	Core Focus / Topic Covered
docs/ARCHITECTURE.md	Overall system architecture, microservices breakdown, and communication protocols.
docs/BUSINESS_INTELLIGENCE_ENGINE.md	Deep dive into the Business DNA profiler, MRI health scoring, and intelligence algorithms.
docs/RULE_ENGINE_SPECIFICATION.md	Specification of rule conditions, signal extractors, and compliance validation rules.
docs/API_CATALOG.md	Exhaustive API catalog detailing request/response schemas and status codes.
docs/KNOWLEDGE_BASE_SPECIFICATION.md	Knowledge base taxonomy and RAG indexing strategy for the AI assistant.
docs/PROJECT_BIBLE.md	Single source of truth for product goals, target MSME personas, and feature roadmaps.
docs/OPERATIONS.md & DEPLOYMENT.md	Production runbooks, container deployment strategies, and CI/CD pipelines.

6. Quality Assurance & Verification Suite

UrsAi incorporates a comprehensive suite of automated verification scripts under /scripts. These scripts execute end-to-end behavioral tests across all development sprints (Sprint 7 through Sprint 16), ensuring strict regression testing and system stability.

- **Sprint 7 Verification:** Validates authentication, business wizard creation, and core REST endpoint contracts.
- **Sprint 8 Verification:** Validates Business DNA calculation, MRI scoring consistency, and SWOT extraction accuracy.
- **Sprint 9 Verification:** Smoke tests end-to-end user workflows, Digital Twin snapshot aggregation, and scenario cloning.
- **Continuous Integration:** Pytest backend test suite (backend/tests) combined with verification scripts validate 100% of API endpoints before release.

End of Specifications Report — Prepared by Antigravity AI Engineering